Modeling and Simulation — Lesson 5
On the Wings of a Butterfly: Exploring Chaos

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The Logistic Map

From Continuous to Discrete Time

- In differential equations, time is a continuous variable t.
- In a **discrete-time model**, time advances in steps: $0, 1, 2, 3, \ldots$
- Such models work well for organisms with **well-defined breeding seasons** (fish, insects), or for phenomena with natural cycles (heartbeats, annual measurements).
- The governing equation is called a **difference equation**:

$$X_{N+1} = f(X_N)$$

- Simplest example: $X_{N+1} = r\,X_N$ (discrete analogue of Malthusian growth).

The Logistic Map

- Consider a population of insects that live one year, lay eggs, and die.
- Resources support a maximum of K individuals.
- Current population X_N uses a fraction $\frac{X_N}{K}$ of resources.
- The unused fraction $1 - \frac{X_N}{K}$ is available for the next generation.
- This gives the **logistic map**:

$$X_{N+1} = r\, X_N \left(1 - \frac{X_N}{K}\right)$$

- Setting $K = 1$ (normalised population):

$$X_{N+1} = r\, X_N(1 - X_N)$$

- The parameter $r \in [0, 4]$ keeps X_N bounded in $[0, 1]$.

Historical Context

- The logistic map was popularised by the biologist **Robert May** in 1976.
- May showed that this deceptively simple equation can produce extraordinarily complex behaviour.
- His paper *"Simple mathematical models with very complicated dynamics"* (Nature, 1976) became one of the most cited papers in mathematical biology.
- The logistic map became a key example in the emerging science of **chaos theory**.

Behaviour for Different Values of r

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- $3.449 \lesssim r \lesssim 3.544$: Population oscillates between **four values** (period-4 cycle).

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- $3 < r \lesssim 3.449$: Population oscillates between **two values** (period-2 cycle).
- $3.449 \lesssim r \lesssim 3.544$: Population oscillates between **four values** (period-4 cycle).
- As r increases further, the period keeps **doubling**: 8, 16, 32, ...
- Beyond $r \approx 3.57$: the system enters **chaos** — the sequence never repeats.
- Within the chaotic regime, there are narrow **windows of periodicity** (e.g. period-3 near $r \approx 3.83$).

The Cobweb Diagram

- A **cobweb diagram** is a graphical tool for visualising the iteration $X_{N+1} = f(X_N)$.
- Plot the parabola $y = r\,x(1-x)$ and the diagonal $y = x$.
- Starting from X_0:
  1. Go vertically to the parabola to find $X_1 = f(X_0)$.
  2. Go horizontally to the diagonal (so X_1 becomes the new input).
  3. Repeat.
- Where the cobweb **spirals inward**, the fixed point is stable.
- Where it **spirals outward** or bounces irregularly, the system is unstable or chaotic.

The Bifurcation Diagram

- The **bifurcation diagram** summarises all possible long-term behaviours as r varies.
- For each value of r, iterate many times to remove transients, then plot the visited values of X.
- The diagram reveals:
  - Single stable branches (fixed points)
  - Period-doubling cascades
  - Dense chaotic bands
  - Windows of order within chaos

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  - Period-doubling cascades
  - Dense chaotic bands
  - Windows of order within chaos
- The ratio between successive bifurcation intervals approaches the **Feigenbaum constant** $\delta \approx 4.66920$.
- This constant is **universal** — it appears in many different systems that undergo period-doubling.

Properties of Chaos

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- **Aperiodicity**: The sequence never exactly repeats. If it did, determinism would force it to repeat forever — making it periodic.
- **Unpredictability**: Despite being deterministic, long-term prediction is practically impossible because of sensitivity to initial conditions.

"Does the flap of a butterfly's wings in Brazil set off a tornado in Texas?"
— Edward Lorenz, 1972

The Lorenz System

From Weather to Chaos

- In the early 1960s, meteorologist **Edward Lorenz** was running a simple computer simulation of atmospheric convection.
- He restarted a run from the middle, rounding an intermediate value from 0.506127 to 0.506.
- The resulting trajectory diverged **completely** from the original.
- This accidental discovery became one of the founding stories of chaos theory.
- Lorenz later asked: *"Does the flap of a butterfly's wings in Brazil set off a tornado in Texas?"*

The Lorenz Equations

The Lorenz system is a simplified model of convection rolls in a heated fluid:

$$\frac{dx}{dt} = \sigma(y - x)$$

$$\frac{dy}{dt} = \rho\, x - y - x\, z$$

$$\frac{dz}{dt} = x\, y - b\, z$$

σ	Prandtl number (ratio of viscosity to thermal diffusivity)
ρ	Rayleigh number (driving force of convection)
b	geometric factor of the convection cell

Classical chaotic parameters: $\sigma = 10,\ \rho = 28,\ b = 8/3$.

The Strange Attractor

- For the classical parameters, all trajectories are eventually attracted to a bounded region of phase space — the **Lorenz attractor**.
- The attractor has a butterfly-like shape with two "wings".
- Trajectories circle one wing, then unpredictably switch to the other, and back again.

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- The attractor has a butterfly-like shape with two "wings".
- Trajectories circle one wing, then unpredictably switch to the other, and back again.
- The attractor is **strange**:
  - It has a **fractal structure** — self-similar at different scales.
  - Nearby trajectories **diverge exponentially**, yet all stay on the attractor.
  - The attractor has **zero volume** but infinite length.

Sensitivity to Initial Conditions

- Start two simulations with initial conditions differing by 10^{-10}.
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- Start two simulations with initial conditions differing by 10^{-10}.
- For a short time, the trajectories look identical.
- After a critical time, they diverge completely and trace different paths around the attractor.
- This is the practical meaning of chaos: **deterministic but unpredictable**.
- The rate of divergence is measured by the **Lyapunov exponent**. A positive Lyapunov exponent is the mathematical signature of chaos.

Why Does This Matter?

- The Lorenz system shows that **simple deterministic rules** can produce behaviour that **looks random**.
- Implications for weather forecasting:
  - Accurate prediction is fundamentally limited in time.
  - Ensemble forecasts (many slightly different initial conditions) are used to estimate uncertainty.

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- Implications for weather forecasting:
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  - Ensemble forecasts (many slightly different initial conditions) are used to estimate uncertainty.
- The same mathematical structure appears in:
  - Laser physics
  - Chemical reactions
  - Population dynamics
  - Electrical circuits
  - Climate science

Fractals: Mandelbrot and Julia Sets

What Is a Fractal?

- A **fractal** is a geometric object that exhibits **self-similarity** at different scales.
- Zooming in reveals structure that resembles the whole.
- Fractals often arise from simple iterative rules applied repeatedly.
- They are intimately connected to chaos: the boundary of the Mandelbrot set, for example, encodes all possible behaviours of a simple quadratic iteration.

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- Examples in nature:
  - Coastlines, mountain ranges
  - Branching of trees, blood vessels, lungs
  - Snowflakes, lightning bolts
  - Romanesco broccoli

Iteration in the Complex Plane

- Recall: a complex number $c = x + iy$ is a point in the 2D plane.
- We study the iteration:

$$z_{n+1} = z_n^2 + c$$

- Starting from $z_0 = 0$, does the sequence $\{z_n\}$ stay bounded or escape to infinity?
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- If $|z_n| > 2$ at any point, the sequence will diverge.
- Two key objects arise from this single equation:
  - **Mandelbrot set**: fix $z_0 = 0$, vary c. Colour by whether c causes divergence.
  - **Julia set**: fix c, vary z_0. Colour by whether z_0 causes divergence.

The Mandelbrot Set

- The Mandelbrot set is the set of all complex numbers c for which the iteration $z_{n+1} = z_n^2 + c$ (starting from $z_0 = 0$) does **not** diverge.
- Points **inside** the set are coloured black.
- Points **outside** are coloured by how quickly they escape (the **escape time**).

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- Points **inside** the set are coloured black.
- Points **outside** are coloured by how quickly they escape (the **escape time**).
- Remarkable properties:
  - The boundary has **infinite complexity** — zooming in reveals ever more detail.
  - Small copies of the whole set appear at all scales.
  - The area is finite, but the boundary has infinite length.
  - The boundary has a fractal dimension ≈ 2.

Julia Sets

- For a fixed c, the **Julia set** is the boundary between initial values z_0 that diverge and those that stay bounded.
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- For a fixed c, the **Julia set** is the boundary between initial values z_0 that diverge and those that stay bounded.
- Each value of c produces a different Julia set.
- There is a deep connection between Mandelbrot and Julia sets:
  - If c is **inside** the Mandelbrot set, the Julia set is a connected, often beautiful fractal.
  - If c is **outside** the Mandelbrot set, the Julia set is a disconnected dust of points (a **Cantor set**).
- The Mandelbrot set is therefore a "map" of all possible Julia sets — it tells us which values of c produce connected fractals.

Connection to Chaos

- The Mandelbrot set and Julia sets arise from the **same iteration** that produces chaos in the logistic map.
- In fact, the logistic map $X_{N+1} = r\,X_N(1 - X_N)$ is related to $z_{n+1} = z_n^2 + c$ by a change of variables.

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- In fact, the logistic map $X_{N+1} = r\,X_N(1 - X_N)$ is related to $z_{n+1} = z_n^2 + c$ by a change of variables.
- The bifurcation diagram of the logistic map corresponds to a slice through the Mandelbrot set along the real axis.
- The Feigenbaum constant $\delta \approx 4.669$ also governs the scaling of structures in the Mandelbrot set.
- All three topics of this lecture — the logistic map, the Lorenz system, and fractals — are manifestations of the same underlying mathematics of **nonlinear dynamics**.

Summary: Chaos in Three Forms

System	Type	Key feature
Logistic map	Discrete, 1D	Period doubling to chaos
Lorenz system	Continuous, 3D	Strange attractor
Mandelbrot set	Complex iteration	Fractal boundary

All three share:

- Sensitivity to initial conditions
- Deterministic yet unpredictable behaviour
- Rich structure from simple rules